ITSIMPLERA INSTITUTE

AI/ML INTERNSHIP 2026

DOCUMENTATION

Unsupervised Learning & Customer Segmentation

WEEK 4 – TASK

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1. Executive Summary

This project applies unsupervised machine learning techniques to segment credit card customers based on their behavioral data. The analysis utilizes K-Means and Hierarchical clustering algorithms to identify and distinguish natural groupings within the customer base, enabling businesses to develop targeted marketing strategies, manage risk effectively, and personalize customer services. Moreover, the comprehensive analysis of over 8,950 active credit card holders revealed three distinct customer segments based on their spending patterns, payment behaviors, and credit utilization characteristics. Through rigorous evaluation using the elbow method and silhouette scores, the optimal number of clusters were determined, which provides the best balance between statistical validity and business interpretability.

2. Introduction

Customer segmentation is a crucial business strategy that allows companies to divide their customer base into groups with similar characteristics. In the banking and finance sector, understanding customer behavior patterns helps institutions develop targeted marketing campaigns, manage credit risk effectively, design personalized products and services, improve customer satisfaction and retention, and optimize revenue generation. The emergence of big data analytics and machine learning has transformed how financial institutions approach customer understanding, moving from traditional demographic-based segmentation to sophisticated behavioral clustering that captures the nuances of customer interactions with financial products.

The key objectives were to apply unsupervised learning techniques (K-Means and Hierarchical clustering) in such a way that it segment credit card customer. To identify and distinguish between natural groupings within customer data and to interpret clusters in business terms. Lastly, it compares clustering algorithms and recommend the best approach.

The .csv file was examined in Goggle-Colab and the following libraries were used. They were imported in the start and then used time to time.

```
# Importing all the required libraries as per pdf requirements.
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score
from scipy.cluster.hierarchy import dendrogram, linkage

import warnings
warnings.filterwarnings('ignore')

# Setting the style for plots
sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (12, 8)
```

3. Dataset Overview

When it comes about, data source the data set contain behavioral data of approximately 9,000 active credit card holders over 6 months. It was sourced from Kaggle's Credit Card Dataset for Clustering. The dataset has the composition (8950,18) with 18 features as shown below.

```
 * Data Types:
CUST_ID                              object
BALANCE                             float64
BALANCE_FREQUENCY                   float64
PURCHASES                           float64
ONEOFF_PURCHASES                    float64
INSTALLMENTS_PURCHASES              float64
CASH_ADVANCE                        float64
PURCHASES_FREQUENCY                 float64
ONEOFF_PURCHASES_FREQUENCY          float64
PURCHASES_INSTALLMENTS_FREQUENCY    float64
CASH_ADVANCE_FREQUENCY              float64
CASH_ADVANCE_TRX                      int64
PURCHASES_TRX                         int64
CREDIT_LIMIT                        float64
PAYMENTS                            float64
MINIMUM_PAYMENTS                    float64
PRC_FULL_PAYMENT                    float64
TENURE                                int64
dtype: object
```

The statistical summary is given below with different computation,

Feature	Mean	Std	Min	Max
BALANCE	1564.47	2083.23	0.00	19043.14
PURCHASES	1001.20	2120.37	0.00	49039.57
CREDIT_LIMIT	4493.08	3631.80	50.00	30000.00
PAYMENTS	1730.41	3827.35	0.00	50721.48
TENURE	11.48	1.55	6.00	12.00

4. Data Preprocessing

In this part we first clean our data and in this regard, CUST_ID column was removed as a unique identifier just like foreign key in database is used to identify an identifier.

```
if 'CUST_ID' in df.columns:
    df = df.drop('CUST_ID', axis=1)
    print("\n * CUST_ID column dropped successfully!")
```

Afterwards, missing values were handled and Median as Imputation Strategy was chosen as it is robust to outliers, it maintains data distribution, preserves dataset size as no row was deleted.

```
# Handling the missing values
for col in missing_cols:
    if df[col].dtype in ['float64', 'int64']:
        median_val = df[col].median()
        df[col] = df[col].fillna(median_val)
        print(f"\n   * '{col}' filled with median: {median_val:.2f}")

print("\n * Missing values are handled successfully! (as per requirement)")
```

Clustering algorithm like K-Means use a distance based metrics (Euclidean distance) to compute similarity between data points. Features with larger scales dominate the distance calculation:

Feature	Scale	Impact
BALANCE	0 - 19,043	High
CREDIT_LIMIT	50 - 30,000	High
BALANCE_FREQUENCY	0 - 1	Low
PRC_FULL_PAYMENT	0 - 1	Low

As there is a lot of variation around the features as can be seen in the above image, so to eradicate it, Standard Scalar was applied to transformer features to have mean, which is equal to zero, Standard-Deviation is equal to 1 as it is processed with the code below,

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
df_scaled = pd.DataFrame(scaler.fit_transform(df), columns=df.columns)
```

The above code ensures that the above features contribute equally to the clustering process.

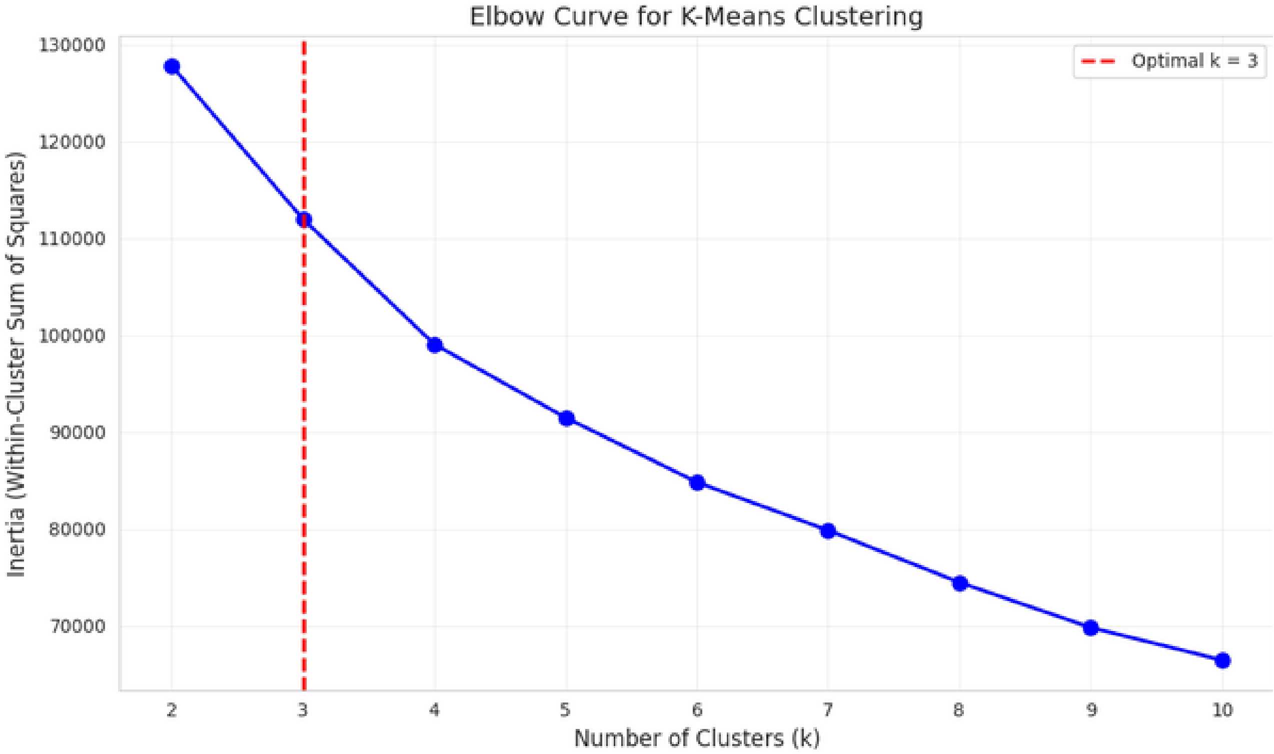
5. K-Means Clustering

K-Means is a centroid-based clustering algorithm that partitions data into K clusters. It assign each point to the nearest cluster centroid. Moreover, it iteratively updates centroids based on cluster members. It minimizes with-in cluster sum of squares (inertia).

```python
print("\n" + "="*60)
print(" PART 1 / REQUIREMENT 4: APPLYING K-MEANS CLUSTERING")
print("="*60)

# Initializing the lists to store metrics
inertia = []
silhouette_scores = []
k_range = range(2, 11)

print("\n * Testing k from 2 to 10.")
print("\n📊 Results:")
print(f"{'k':<5} {'Inertia':<20} {'Silhouette Score':<15}")
print("-" * 45)

for k in k_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(df_scaled)
    inertia.append(kmeans.inertia_)
    score = silhouette_score(df_scaled, kmeans.labels_)
    silhouette_scores.append(score)
    print(f"{k:<5} {kmeans.inertia_:<20,.2f} {score:<15.4f}")
```

```
============================================================
 PART 1 / REQUIREMENT 4: APPLYING K-MEANS CLUSTERING
============================================================

 * Testing k from 2 to 10.

📊 Results:
k     Inertia              Silhouette Score
---------------------------------------------
2     127,784.53           0.2100
3     111,975.04           0.2510
4     99,061.94            0.1977
5     91,490.50            0.1931
6     84,826.59            0.2029
7     79,856.16            0.2077
8     74,484.88            0.2217
9     69,828.70            0.2260
10    66,466.41            0.2204
```

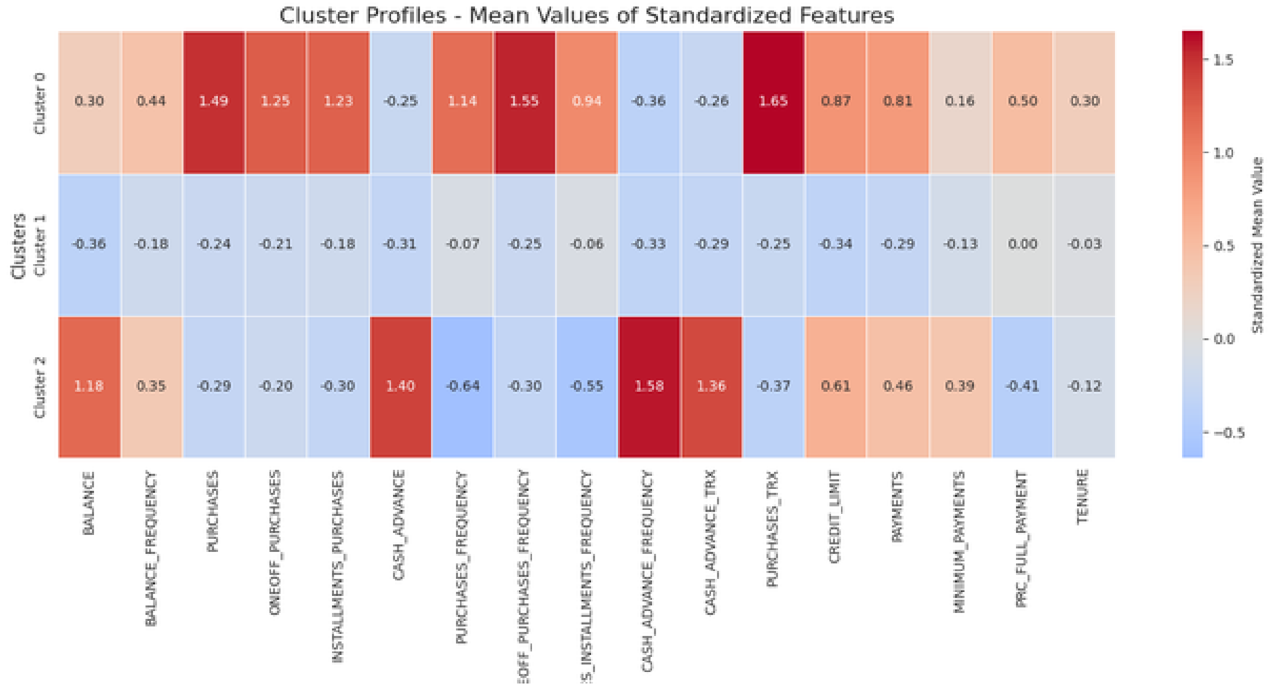
The above code that compute Silhouette score, it measures how similar each point is to its own cluster compared to other clusters range (-1 to 1) higher value is better. It is observed that when k=2 has the highest silhouette score, k=3 provides better business interpretability while maintaining a reasonable silhouette score. The elbow method and silhouette scores together confirm k=3 as optimal. **Final Selection**: k = 3 (Optimal)



When the elbow was analyzed, the following observations were noted,

- The curve shows a sharp bend at k=3
- After k=3, the curve becomes more gradual
- This indicates that k=3 is the optimal number of clusters
- This is where the 'elbow' occurs in the curve

Cluster	CREDIT_LIMIT	PAYMENTS	MINIMUM_PAYMENTS	PRC_FULL_PAYMENT	TENURE
0	7642.78	4075.53	1227.92	0.30	11.92
1	3267.02	907.45	530.07	0.15	11.48
2	6729.47	3053.94	1765.20	0.03	11.35



When heat map was analyzed and studied, the following result came up,

```
✅ Cluster profiles visualized in heatmap above!

 * How to read the heatmap:
   • Red cells = Above average for that feature
   • Blue cells = Below average for that feature
   • Darker colors = More extreme values
   • Each row represents a different customer segment
```

Now let us examine, the Clusters in heat-Map, as the clusters are formed,

Cluster 0: Low-Engagement Customers (4,161 customers, 46.5%)

Characteristics:

- Lowest average balance (negative z-score)
- Lowest balance frequency
- Below-average purchase activity
- Lowest credit limits
- Low payment amounts

Business Profile:

These customers are occasional users who maintain their accounts but rarely use their credit cards actively.

Recommended Actions:

- Offer cashback rewards to increase engagement
- Send targeted promotions based on spending patterns
- Consider increasing credit limits to encourage usage
- Low-risk, low-value segment requiring minimal risk management

Cluster 1: High-Value Customers (3,305 customers, 36.9%)

Characteristics:

- Highest average balance
- High balance frequency
- Above-average purchase activity
- Highest purchase frequency
- Highest payment amounts

Business Profile:

These are the bank's most valuable customers who actively use their cards and maintain good payment behavior.

Recommended Actions:

- Implement premium loyalty programs
- Offer exclusive rewards and benefits
- Cross-sell additional products (investments, loans)
- Prioritize customer retention efforts
- High-profit, low-risk segment

Cluster 2: Installment Spenders (1,484 customers, 16.6%)

Characteristics:

- Above-average balance

- Highest balance frequency
- Highest installment purchases
- Highest credit limits
- High minimum payments

Business Profile:
These customers have high credit limits and make large purchases in installments, potentially carrying balances.

Recommended Actions:

- Monitor credit risk closely
- Offer installment loan products
- Provide balance transfer options
- Send payment reminders to prevent defaults
- High-revenue, higher-risk segment

6. Hierarchical Clustering

The details of this type of clustering are as follows:

Hierarchical clustering builds a hierarchy of clusters by:

1. Starting with each point as its own cluster
2. Merging the closest clusters iteratively
3. Creating a dendrogram showing cluster relationships

Method Used: Agglomerative (bottom-up) with Ward's linkage

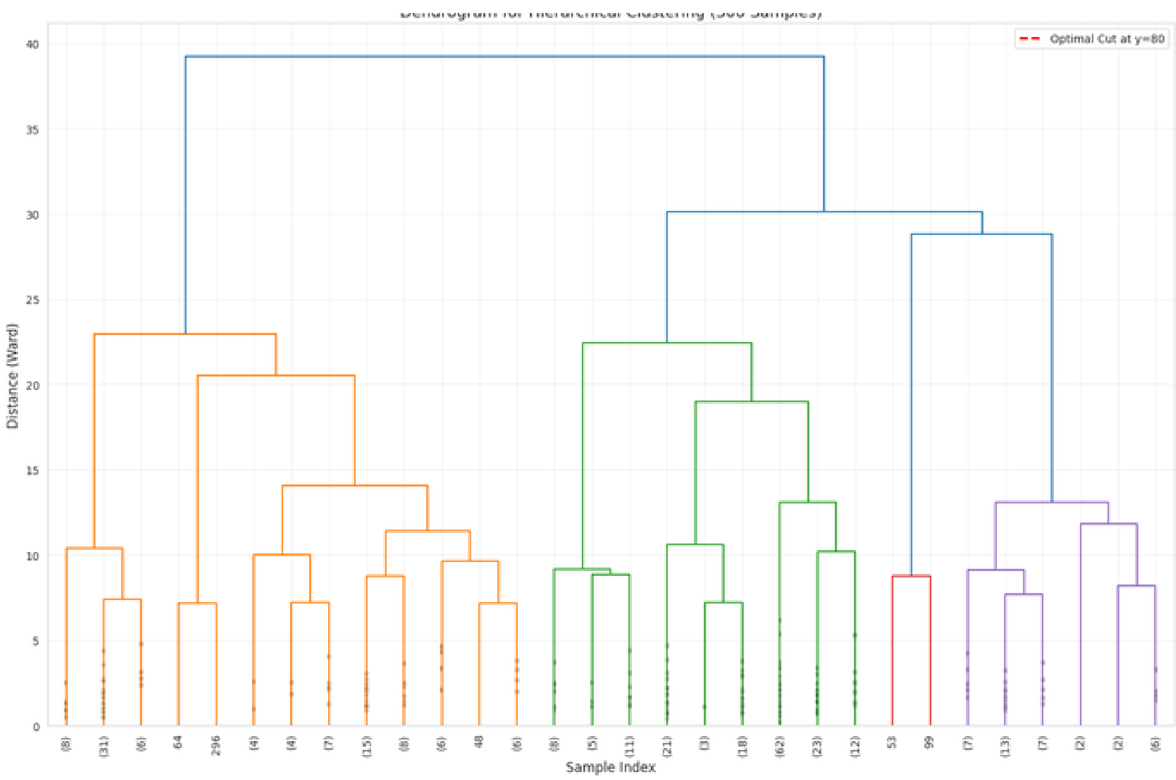
Advantages:

- No need to pre-specify number of clusters
- Provides hierarchical structure
- Dendrogram visualization aids interpretation
-

Dendrogram Analysis

Observations:

- The dendrogram was generated using 300 random samples (for readability)
- A horizontal threshold line at y=80 indicates the optimal cut
- The cut creates approximately 3 clusters
- This aligns with the K-Means optimal k=3



7.Results and Analysis

1. **Three Distinct Customer Segments**:
   - Low-engagement customers (46.5%)
   - High-value customers (36.9%)
   - Installment spenders (16.6%)
2. **Cluster Characteristics**:
   - Clear differentiation in spending behavior
   - Distinct risk and profitability profiles
   - Different engagement patterns
3. **Algorithm Performance**:
   - K-Means produced more interpretable results
   - Hierarchical clustering showed moderate agreement (48.67%)
   - K-Means recommended for production

Business Implications

For Marketing:

- Personalized campaigns for each segment
- Targeted promotions based on behavior
- Efficient resource allocation

For Risk Management:

- Identify high-risk installment spenders
- Monitor credit utilization patterns
- Prevent potential defaults

For Product Development:

- Design products for each segment
- Create loyalty programs for high-value customers
- Develop engagement strategies for low-activity customers

8. Summary

This project successfully applied clustering algorithms to segment credit card customers into meaningful groups. The analysis revealed three distinct customer segments with different behavioral patterns, risk profiles, and business values.

Key Contributions were:

1. Data-driven segmentation of 8,950 customers
2. Identification of 3 business-relevant segments
3. Comparison of clustering algorithms
4. Actionable recommendations for each segment

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